Stochastic Audit of Electricity Demand in Systems with Telemetric Censorship: A Remote Sensing and Photometric Calibration Approach

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***Abstract*—The performance evaluation of large-scale power systems frequently faces institutional opacity and the censorship of direct telemetric measurements. This article presents a bidirectional methodological framework that integrates stochastic modeling and spatial remote sensing to audit served demand under data scarcity. In the first phase, the sensitivity of the Type I Error of the Analysis of Variance (ANOVA) to regional heteroscedasticity and sample imbalance is evaluated through Monte Carlo simulations (5000 iterations). In the second phase, a nonlinear cross-calibration algorithm based on photometric power ratios is designed, utilizing multispectral time series from the DMSP and NOAA-VIIRS sensors. The stochastic audit confirms extreme asymmetry and heteroscedasticity ($p = 3.90 \times 10^{-18}$). The model estimates the maximum served demand of the National Electric System for the 2021-2026 period at $\hat{\mu} = 6315.42$ MW (99% CI). Through cross-correlation tests ($r > 0.73$) and statistical power evaluation, the consistency of the official operational threshold of 14000 MW is refuted, analytically demonstrating that the grid's instability is a consequence of a systemic degradation in thermal inertia rather than excessive consumption.**

***Index Terms*—Data Censorship, Energy Audit, Nighttime Lights (NTL), Parametric Robustness, Stochastic Inference.**

I. Introduction

THE performance evaluation of large-scale industrial systems requires precise measurements of operational variables. However, in environments characterized by institutional opacity, access to telemetric logs is often censored. In such scenarios, computational mathematics offers robust tools for inferring latent variables through physical proxies, such as nighttime light (NTL) emission.

Fundamental studies, such as the one conducted by Zhang, Li, and Chen (2020) [1], validated the VIIRS series to quantify the contraction of Venezuelan economic activity, reporting a loss of over 30% in the sum of urban light using trend filters. This research extends this methodological paradigm by surpassing the traditional descriptive approach and implementing a comprehensive stochastic audit.

This work provides three fundamental innovations: 1) the validation of regional parametric assumptions through formal heteroscedasticity tests; 2) the quantification of bias in linear inferential models via Monte Carlo simulations; and 3) the integration of geospatial data up to the year 2026 within a hybrid processing architecture (Google Earth Engine and R) to determine the actual served demand.

II. Design of the Stochastic Analysis

To ensure analytical rigor, the research circumvents telemetric censorship by defining a testable parametric framework.

A. Hypothesis Formulation and the Westinghouse Theorem

The Null Hypothesis (H_0) establishes the official operational limit ($\mu_{\text{demand}} \geq 14000$ MW), while the Alternative Hypothesis (H_a) assumes a drastic contraction ($\mu_{\text{demand}} < 14000$ MW). The classical analysis of diversified maximum demand is governed by the Westinghouse theorem. Given the censorship of micro-parametric variables (active subscribers, industrial plant factors), this study employs Nighttime Satellite Photometry as a macroscopic observable integrator. The radiance captured by satellite sensors represents the final physical aggregation of the diversified load served across geographical nodes.

B. Monte Carlo Simulation and ANOVA Robustness

To validate statistical tools in unstable systems, the sensitivity of the Fisher-Snedecor F-statistic was evaluated through 5000 Monte Carlo iterations. The simulations demonstrated that regional heteroscedasticity—common in grids experiencing asymmetrical rationing—increases the Type I Error by more than 300%. This invalidates classical parametric approaches and necessitates a nonlinear stochastic calibration for interpreting satellite data.

III. Geospatial Acquisition and Calibration

Data processing was structured across two computational layers. The acquisition layer utilized Python and Google Earth Engine [2] to ingest data from the DMSP-OLS (historical) and NOAA-VIIRS (contemporary) missions [3].

Three critical nodes were isolated: Carabobo (industrial), Miranda (capital), and Zulia (western). To compensate for the spatial and temporal resolution discrepancies between the two sensors, a weighted marginal transfer constant ($\kappa = 0.002988$) was calculated. This constant was designed to harmonize physical radiance ($\text{nW} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$) with historical equivalents of served megawatts. The inference layer was subsequently developed in R to compute statistical power and perform the contrast of means.

IV. Results and Discussion

A. Unbiased Estimation of Served Demand

The processing of 65 monthly observations of contemporary spatial telemetry demonstrated that the grid operates under a chronic asymmetric regime ($S_k = 0.505$) and extreme heteroscedasticity (Levene, $p = 3.90 \times 10^{-18}$). Applying the κ coefficient, the stochastic mean of the current maximum demand stands at 6315.42 MW, rigorously bounded by a 99% confidence interval of $[6019.52\ ;\ 6611.31]$ MW.

B. Magnitude of Discrepancy and Stochastic Synchronization

The empirical distancing from the institutional operational limit yields an absolute effect size (Cohen's d) of $|d| = 6.74$ [4]. This divergence of over six standard deviations statistically confirms that the official value belongs to a mutually exclusive population. The inferential evaluation ($n = 65$, $\alpha = 0.01$) guarantees an empirical statistical power ($1 - \beta$) of 100%.

To rule out the hypothesis of isolated local distribution failures, a Pearson cross-correlation matrix was computed. The nodes exhibited a high level of coupling ($r_{\text{Carabobo, Miranda}} = 0.835$; $r_{\text{Zulia, Carabobo}} = 0.738$). This global correlation (> 0.73) proves that the system operates as a monolithic block in deficit, validating that the observed light oscillations are the result of centralized under-frequency load shedding maneuvers.

C. Systemic Stability and Grid Dynamics

Confirming a served demand of approximately 6300 MW allows for a reinterpretation of the instabilities observed in the 765 kV trunk transmission ring [5]. By operating with a real load significantly lower than its nominal design, the infrastructure experiences the Ferranti Effect, generating capacitive overvoltages. Furthermore, the intermittent oscillations serve as an unequivocal symptom of a critical loss of rotational inertia within the base thermoelectric park, inducing angular collapse in response to minimal variations in hydraulic dispatch.

V. Conclusion

This study comprehensively demonstrates that nighttime spatial radiance serves as a highly reliable algorithmic proxy for load estimation in electrically unstable and telemetrically opaque systems. The developed computational model analytically and irrefutably refutes the institutional hypothesis of excessive consumption. The statistical probability of the grid supporting the declared 14000 MW is zero. The chronicity of the blackouts and the voltage instability are ultimately defined as the physical response of a transmission system operating with degraded thermal inertia under a low-load regime (Ferranti Effect). The proposed methodology consolidates geospatial stochastic inference as a transparent and robust tool for independent technical auditing.

References

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